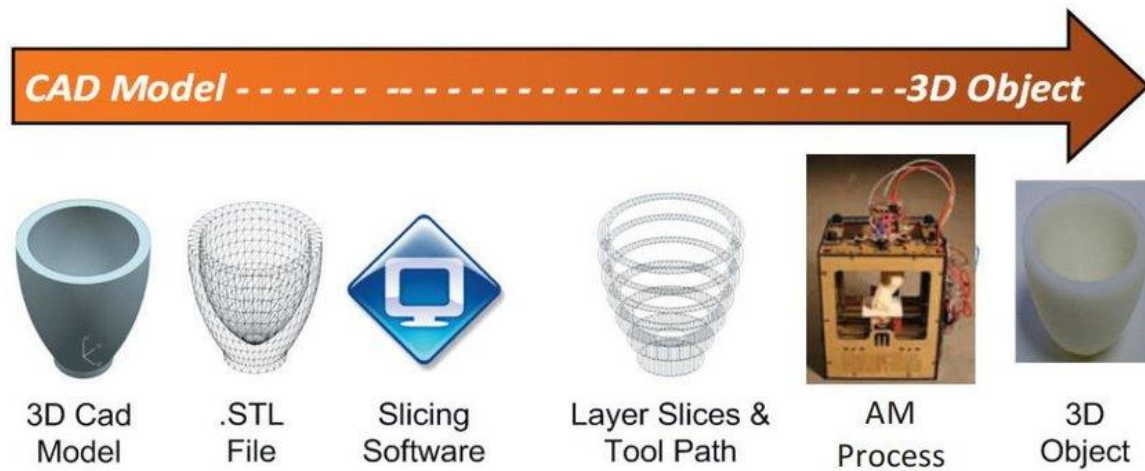


INTRODUCTION TO ADDITIVE MANUFACTURING



COURSE CONTENT

- 1.1.1 Introduction
- 1.1.2 Additive Manufacturing
- 1.1.3 Steps in Additive Manufacturing
- 1.1.4 Classification of AM Processes
- 1.1.5 AM in New Product Development
- 1.1.6 Materials used in AM
- 1.1.7 Additive v/s Conventional Manufacturing processes
- 1.1.8 Applications of AM in Various Domains

Prof. Shiva Krishna J

Dept. of Mechanical & Automation Engg., Agnel Institute of Technology and Design, Assagao

1.1.1 INTRODUCTION

A prototype is a first or preliminary version of a device, system, or vehicle from which subsequent forms are developed or refined.

Unlike the conventional view that a prototype must be a physical object, the term broadly covers all types of representations used in the product development process. These can include mathematical models, pencil sketches, virtual models, foam models, 3D-printed parts, or fully functional physical approximations of the product.

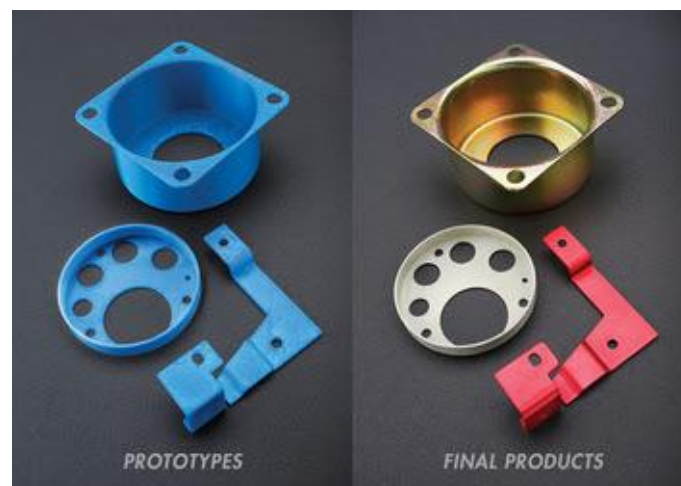
Prototyping is the process of creating these representations. It can range from running a simulation on a computer program to building a working physical model that mirrors the final product.

ROLE OF PROTOTYPES IN PRODUCT DEVELOPMENT

Prototypes play multiple critical roles in product design and development, such as:

1. Experimentation and Learning: Exploring design ideas, testing feasibility, and identifying potential flaws at an early stage.
2. Testing and Proofing: Validating performance, safety, ergonomics, and durability before large-scale production.
3. Communication and Interaction: Helping designers, engineers, clients, and stakeholders visualize the product and provide constructive feedback.
4. Synthesis and Integration: Bringing together mechanical, electrical, software, and user-interface aspects into a unified model.
5. Scheduling and Milestones: Serving as tangible checkpoints in the development timeline to assess progress and readiness for subsequent phases.
6. Risk Reduction: Minimizing the cost and impact of errors by identifying and resolving issues before mass production.
7. Innovation Support: Allowing room for iterative improvements and encouraging creative exploration.

PROTOTYPE v/s FINAL PRODUCT



Difference between a prototype and Final Product

Although prototypes serve as the foundation for the final product, the creation of a prototype differs in several fundamental ways:

1. Material:

- Final products are often made with durable, high-quality, and sometimes costly materials.
- Prototypes, on the other hand, may use substitute or easily machinable materials to reduce costs and speed up development.
- In some cases, final materials may not even be available yet during the prototyping stage.

2. Process:

- Final products typically use mass-production processes (e.g., injection molding, die casting, automated assembly).
- Prototypes are usually fabricated using rapid or low-volume methods such as CNC machining, 3D printing, laser cutting, or handcrafting.
- This often results in visual or structural differences between the prototype and the final product.

3. Verification and Testing:

- Final products undergo rigorous quality assurance processes, including statistical sampling, regulatory compliance checks, and durability testing.
- Prototypes, however, are usually inspected individually, with the expectation of modifications, rework, or redesign. Some compliance requirements may be waived during this stage.

4. Cost and Time:

- The final product aims for cost-effectiveness in large-scale manufacturing.
- Prototypes may be more expensive per unit due to specialized fabrication and one-off tooling, but they save overall costs by preventing large-scale errors.

5. Functionality:

- Prototypes may be partial or full representations of the final product.
- Types include visual prototypes (appearance only), functional prototypes (performance testing), and concept prototypes (early idea validation).
- The final product integrates all required functionalities with consistency, reliability, and scalability.

Engineers and prototype specialists strive to minimize the gap between a prototype and the final product. For example:

- If the same material cannot be used, substitutes with similar physical or mechanical properties are chosen.
- If mass-production tooling is unavailable, rapid prototyping technologies simulate the required precision and performance.
- Iterative prototyping cycles help progressively refine the product until it closely matches the intended final design.

PHASES OF PROTOTYPING

- Prototyping or model making is an age-old practice, rooted in the need to transform conceptual designs into tangible forms. A prototype serves as a bridge between design and production, enabling visualization, testing, and refinement before mass manufacturing.
- Traditionally, prototypes have been fabricated through processes such as material removal, casting, moulding, and adhesive joining, using a variety of materials including aluminium, zinc, urethanes, plastics, and wood.
- Over time, prototyping has evolved through three distinct phases, the latter two driven by advancements in digital technology and automation. Much like the evolution of computer graphics, physical prototyping has now entered its third and most advanced phase.

1. First Phase: Manual Prototyping:

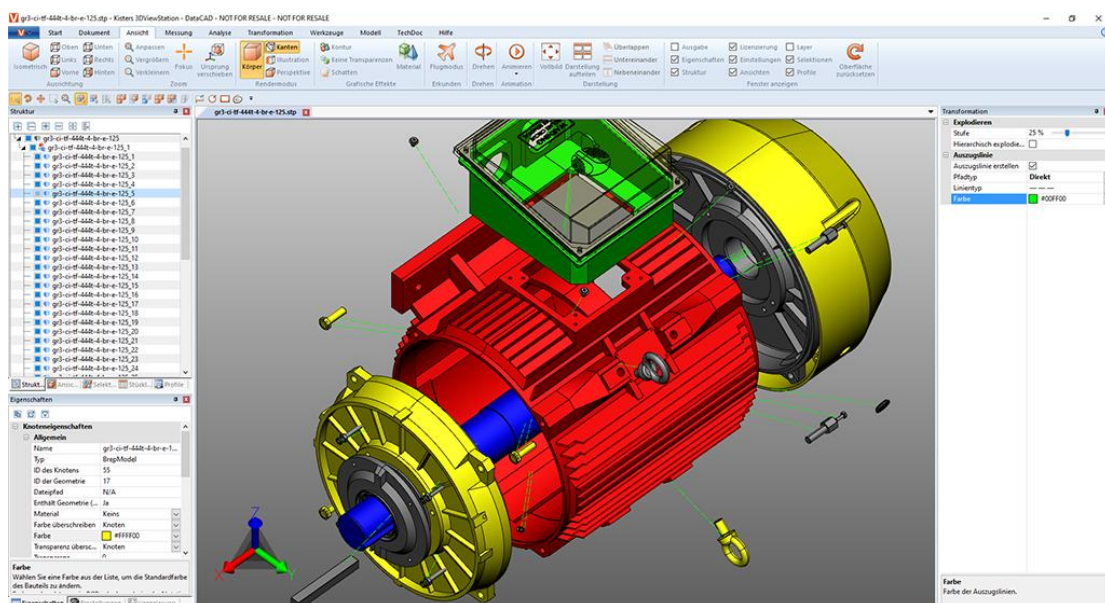
- Manual prototyping began centuries ago as humans crafted tools and products to meet their needs.
- In this phase, prototypes were often simple, handmade, and craft-based, relying heavily on the skill of artisans.
- Fabrication was time-consuming and labour-intensive, typically taking around four weeks for a moderately complex prototype.
- Techniques included hand carving, wood/metal shaping, and basic casting methods.

Limitations:

- Low accuracy and repeatability.
- Difficulties in scaling up to mass production.
- Long fabrication cycles.

2. Second Phase: Soft (or) Virtual Prototyping:

- With the rise of CAD (Computer-Aided Design), CAE (Computer-Aided Engineering), and CAM (Computer-Aided Manufacturing) in the early 1980s, virtual prototyping emerged.



Virtual Prototyping

- Designers could now create digital models, simulate stress, strain, thermal effects, and other behaviours by specifying precise material properties.
- Fabrication of physical models still relied on traditional/manual methods, taking up to 16 weeks despite improvements like CNC machining.

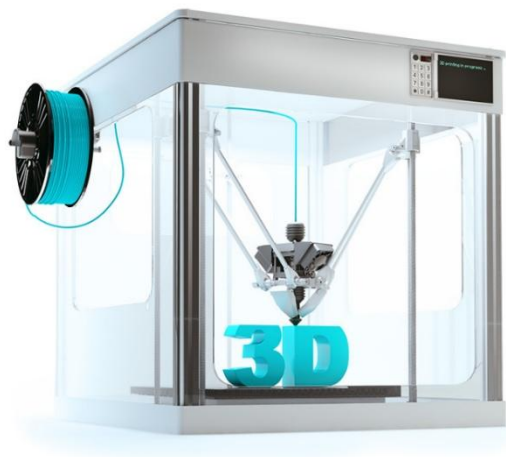
Advantages:

- Multiple design iterations could be performed quickly by altering parameters in the software.
- Complex designs nearly twice as complex as before became feasible.
- Allowed early detection of flaws without physical fabrication.

Limitations:

- Virtual prototypes cannot anticipate or simulate all real-world phenomena. Unmodeled effects may cause performance gaps in actual usage.

3. Third Phase: Rapid Prototyping:



Rapid Prototyping

- The late 1980s marked a revolutionary leap with the advent of Rapid Prototyping (RP), also known as Solid Freeform Fabrication (SFF), Layer Manufacturing Technology, Additive Manufacturing (AM) and Desktop Manufacturing
- Uses layer-by-layer fabrication directly from CAD models.
- Eliminates tooling, moulds, or machining setups, drastically reducing lead time.
- Capable of producing prototypes with 3× the complexity of 1970s designs in as little as 3 weeks.
- Since 1988, more than 20 different RP techniques have been developed (e.g., Stereolithography, Selective Laser Sintering, Fused Deposition Modelling, Laminated Object Manufacturing).

Advantages:

- Rapid turnaround for complex parts.
- High accuracy and repeatability.
- Supports a wide range of materials (plastics, metals, ceramics, composites).
- Enables functional testing (fit, form, performance).

1.1.2 ADDITIVE MANUFACTURING

Additive Manufacturing (AM), commonly known as 3D Printing, is the layer-by-layer fabrication of three-dimensional physical models directly from CAD data.

- It represents a group of advanced techniques that enable the rapid and accurate creation of parts, assemblies, or functional prototypes directly from digital models.
- Unlike traditional subtractive or formative manufacturing methods, AM builds objects by adding material only where needed, making it an efficient, flexible, and innovative approach to product development and production.

KEY FEATURES AND ADVANTAGES OF ADDITIVE MANUFACTURING

1. Layer-by-Layer Fabrication:

- AM constructs objects by stacking thin cross-sections generated from CAD data.
- This approach allows the creation of complex geometries, including lattice structures, internal cavities, thin walls, and *parts within parts*, which are extremely difficult or impossible to achieve using conventional methods.

2. Simplified Construction Process:

- A 3D model is digitally sliced into layers, which are then printed consecutively.
- Example: Instead of machining a sphere, AM builds it through successive layers of circular slices, achieving precision with minimal effort.

3. Reduced Human Intervention:

- Once the CAD design is prepared and the machine is set up, AM systems usually operate unattended until the build is complete.
- Post-processing (e.g., support removal, surface finishing, curing) is required, but operator involvement is much lower than in conventional manufacturing.

4. Time and Cost Efficiency:

- Prototypes and functional parts can be produced in hours instead of weeks.
- Eliminates the need for expensive tooling, moulds, and complex setups.
- Materials used in AM are often more economical, especially for prototyping and low-volume production.

5. Accessibility and Flexibility:

- AM machines range from desktop-scale printers (suitable for offices and classrooms) to industrial-grade systems for aerospace, automotive, and biomedical industries.
- Designers can quickly modify and reprint parts, speeding up the design–test–iterate cycle.

6. Functional Prototyping and End-Use Parts:

- AM produces functional prototypes that can be tested for fit, form, and performance.
- Increasingly, AM is being used to manufacture end-use parts, such as lightweight aerospace components, medical implants, and customized consumer products.

7. Enables Innovation:

- Provides unparalleled design freedom by removing traditional manufacturing constraints.
- Supports topology optimization, lattice structures, and multi-material printing.
- Encourages sustainability by reducing material waste and enabling localized, on-demand production.

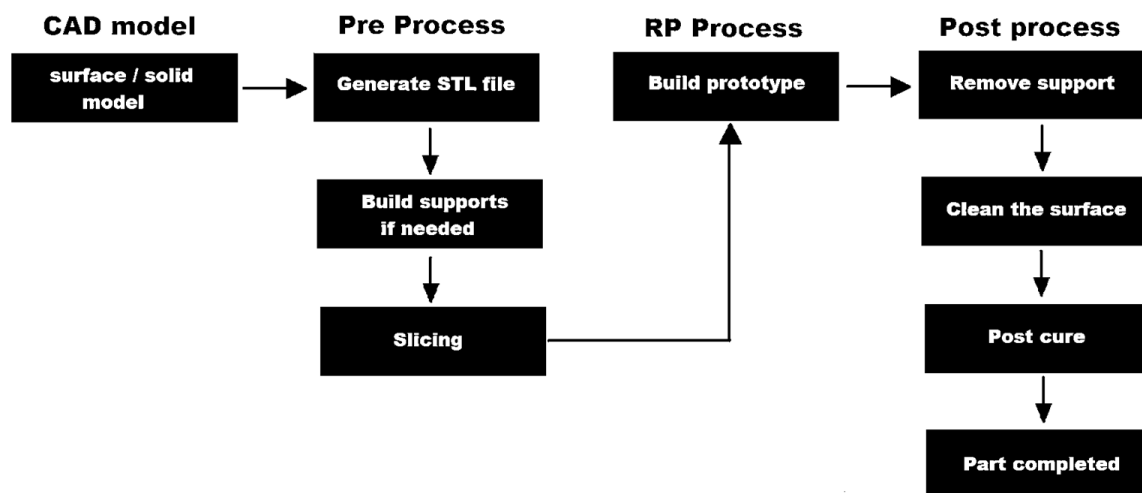
COMMONLY USED TERMS FOR ADDITIVE MANUFACTURING

Additive Manufacturing is also referred to by several related terms:

- 3D Printing
- Direct CAD Manufacturing
- Desktop Manufacturing
- Instant Manufacturing
- CAD-Oriented Manufacturing
- Layer Manufacturing
- Material Deposition Manufacturing
- Material Addition Manufacturing
- Solid Freeform Fabrication (SFF)

1.1.3 STEPS IN ADDITIVE MANUFACTURING

AM in its basic form can be described as the production of 3D parts from CAD data within a reduced time scale. The basic methodology of all AM processes can be summarized as shown in the following steps:



Steps in Additive Manufacturing

1. Development of a CAD Model:

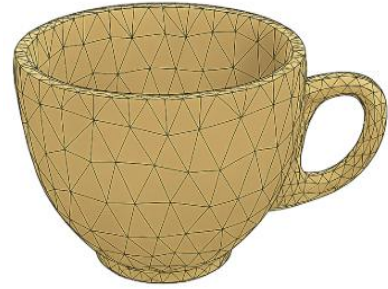
The process begins with the generation of a CAD model of the desired object, which can be done by one of the following ways:

- Conversion of an existing two-dimensional (2D) drawing
- Importing scanned point data into a CAD package
- Creating a new part in CAD using various solid modelling packages
- Altering an existing CAD model



2. Generation of STL File:

The developed 3D CAD model is tessellated and converted into Standard Triangulation Language (STL) files that are required for AM processes. Tessellation is a piecewise approximation of surfaces of a 3D CAD model using a series of triangles. The STL file represents the surface of the model as an array of triangles and consists of the X, Y, and Z coordinates of the three vertices of each triangle, along with an index that describes the orientation of the surface normal.



3. Build Support Structures:

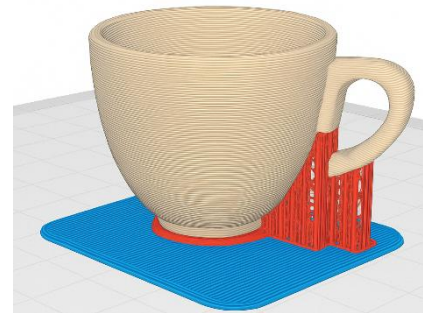
Since parts are built layer by layer, some areas may be unsupported or overhanging, which could cause distortion. Therefore, certain AM processes require support structures or a base to be added to the file. These supports are fabricated along with the model and later removed.

4. Slicing:

Slicing involves creating contours of the geometry at various heights according to the chosen layer thickness. Once the STL file has been generated from the original CAD data, the next step is to slice the object to create a slice file (SLI).

- Thicker slices → faster build, but rougher surface finish.
- Thinner slices → slower build, but finer detail.

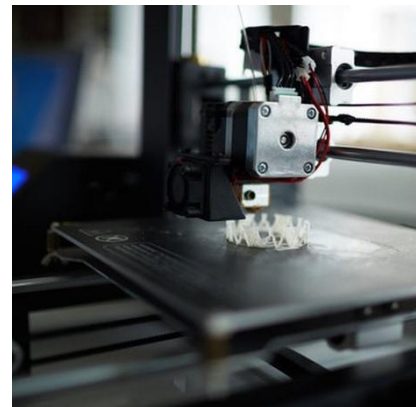
The sliced files are then merged into a final build file.



5. Building the Part:

The AM process is additive, i.e., the part is built layer by layer from the bottom up. Each new layer is automatically bonded to the previous one. Depending on the material and technology, bonding can be achieved through:

- Ultraviolet (UV) light
- Carbon dioxide lasers
- Heat-sensitive adhesives
- Melting and solidification of the material itself



6. Post-Processing:

After the build, the part is removed from the machine and subjected to post-processing. This may include:

- Removing support structures
- Curing or sintering for improved strength
- Filling voids
- Surface finishing (sanding, polishing, coating, painting)

The level of post-processing depends on the final application and quality requirements of the part.

1.1.4 CLASSIFICATION OF ADDITIVE MANUFACTURING PROCESSES

AM processes can be classified in multiple ways depending on the initial form of the build material and the underlying process mechanism.

I. BASED ON BUILD MATERIALS:

1. Liquid-based Systems:

- In these systems, the starting material is in liquid form and solidifies through curing (by light, heat, or chemical processes).
- The following AM systems fall into this category:
 - Stereolithography Apparatus (SLA)
 - Solid Ground Curing (SGC)
 - Digital Light Processing (DLP)
 - Masked Stereolithography (MSLA)
 - Lithography-based Metal Manufacturing (LMM)

2. Solid-based Systems:

- The material is in solid form such as wire, sheets, laminates, or pellets.
- The process may involve cutting and bonding/joining or melting and fusing.
- The following AM systems fall into this definition:
 - Laminated Object Manufacturing (LOM)
 - Fused Deposition Modelling (FDM)
 - Paper Lamination Technology (PLT)
 - Multi-Jet Modelling (MJM)
 - Selective Lamination Composite Object Manufacturing (SLCOM)

3. Powder-based Systems:

- Material is in the form of fine powder particles (metals, polymers, or ceramics).
- Parts are formed by binding or fusing the powder, using a laser, electron beam, or binder solution.
- The following AM systems fall into this group:
 - Selective Laser Sintering (SLS)
 - Direct Metal Laser Sintering (DMLS)
 - Three-Dimensional Printing (3DP)
 - Laser Engineered Net Shaping (LENS)
 - Electron Beam Melting (EBM)
 - Direct Metal Deposition (DMD)

II. BASED ON PROCESS MECHANISM (ISO/ASTM Standard Classification):

1. Vat Photopolymerization:

- Process where liquid photopolymer is selectively cured by a light source.
- Examples:
 - Stereolithography (SLA)
 - Digital Light Processing (DLP)
 - Masked Stereolithography (MSLA)
 - Lithography-based Metal Manufacturing (LMM)

2. Material Extrusion:

- Material is selectively dispensed through a nozzle or orifice.
- Example:
 - Fused Deposition Modelling (FDM)

3. Powder Bed Fusion:

- Thermal energy selectively fuses regions of a powder bed.
- Examples:
 - Selective Laser Sintering (SLS)
 - Direct Metal Laser Sintering (DMLS)
 - Electron Beam Melting (EBM)
 - Multi Jet Fusion (MJF)

4. Material Jetting:

- Droplets of build material are selectively deposited.
- Examples:
 - Material Jetting (MJ)
 - Nanoparticle Jetting (NPJ)

5. Directed Energy Deposition:

- Focused thermal energy melts material as it is being deposited.
- Examples:
 - Laser Engineered Net Shaping (LENS)
 - Electron Beam Additive Manufacturing (EBAM)
 - Direct Metal Deposition (DMD)

6. Sheet Lamination:

- Sheets of material are bonded together layer by layer.
- Examples:
 - Laminated Object Manufacturing (LOM)
 - Selective Lamination Composite Object Manufacturing (SLCOM)
 - Plastic Sheet Lamination (PSL)

Summary of popular AM processes with Material used and Application

Category	Process	Material Used	Applications
<i>Liquid-based</i>	Stereolithography (SLA)	Photopolymer resin	Dental, Jewellery, Medical Devices, Prototypes
	Material Jetting (PolyJet, MJM)	Liquid photopolymers	Multi-colour prototypes, Medical models, Realistic parts
<i>Solid-based</i>	Fused Deposition Modelling (FDM)	Thermoplastic filament (PLA, ABS, Nylon)	Prototyping, Jigs & Fixtures, Low-cost models
	Laminated Object Manufacturing (LOM)	Sheets of paper, plastic	Visual models, Design studies

Powder-based	Selective Laser Sintering (SLS)	Polymer powders (Nylon, TPU)	Functional prototypes, Automotive & Aerospace parts
	Direct Metal Laser Sintering (DMLS)	Metal powders (Steel, Ti, Al, Inconel)	Aerospace, Medical implants, Tooling
	Binder Jetting	Sand / metal powders	Casting moulds, Full-colour models, Metal parts
	Laser Engineered Net Shaping (LENS)	Metal powder	Repair of Turbine blades, Large aerospace parts

1.1.5 ADDITIVE MANUFACTURING IN NEW PRODUCT DEVELOPMENT

Product is just an idea before its actual development. Product development typically refers to all stages involved in bringing a product from concept or idea through market release and beyond. In other words, the product development process incorporates a product's entire journey.



Flowchart shows the steps in New Product Development

AM supports the New Product Development cycle from idea generation to market launch by providing speed, cost savings, design freedom, and flexibility. It is especially valuable in industries requiring customization, lightweighting, and rapid innovation.

1. Idea Generation:

- The first stage of New Product Design where innovative product concepts are conceived.
- Market trends, customer needs, and technological opportunities are analysed.
- AM helps here by enabling quick visualization and even rough early prototypes to communicate ideas more effectively.

2. CAD Design:

- The product idea is converted into a 3D CAD model.
- Designers focus on form, function, and aesthetics.
- AM encourages design freedom, since constraints of traditional manufacturing (e.g., machining, casting) are reduced. Lightweight structures and internal channels can be designed without concern for tooling limitations.

3. Prototype Development:

- Turning concepts into actual product designs with engineering specifications, prototypes, and models.
- Unlike traditional methods, in AM no moulds, dies, or complex setups are required.

- AM allows low-cost prototyping, even for complex geometries.
- Functional (or) aesthetic prototypes are produced quickly

4. Testing & Validation:

- Prototypes are subjected to functional tests (strength, ergonomics, thermal, aerodynamic, etc.).
- Fit and assembly are checked with other components.
- AM enables early validation, reducing chances of late-stage design failures.
- Multiple prototype versions can be tested in parallel.

5. Design Iteration:

- Based on testing feedback, design changes are incorporated.
- AM makes rapid iteration possible: quick redesigns, reprinting, and testing.
- Iteration cycles become shorter, improving the overall design quality.

6. Final Production:

- Once optimized, the product moves to production.
- For mass production, conventional methods such as casting, injection moulding, machining may be used.
- For low-volume, complex, or customized products, AM itself can serve as the final production method (e.g., aerospace parts, dental implants).
- Hybrid approaches (AM + machining) are also sometimes used

7. Market Launch:

- Full-scale production, marketing, and distribution of the new product into the market.
- With faster prototyping and reduced design cycles, time-to-market is significantly reduced.
- Early product testing with real users becomes possible, improving acceptance.
- Companies gain a competitive advantage by launching innovative products ahead of competitors.

1.1.6 MATERIALS IN ADDITIVE MANUFACTURING

Additive Manufacturing makes use of a wide range of materials depending on the process, application, and performance requirements. These materials can be broadly classified into Polymers, Metals, Ceramics, Composites, and Biomaterials.

Each class is available in specific forms (filaments, powders, resins, sheets, etc.) and is compatible with certain Additive Manufacturing processes.

1. Polymers: Polymers are the most common entry point into AM due to low cost, ease of processing, and versatility.

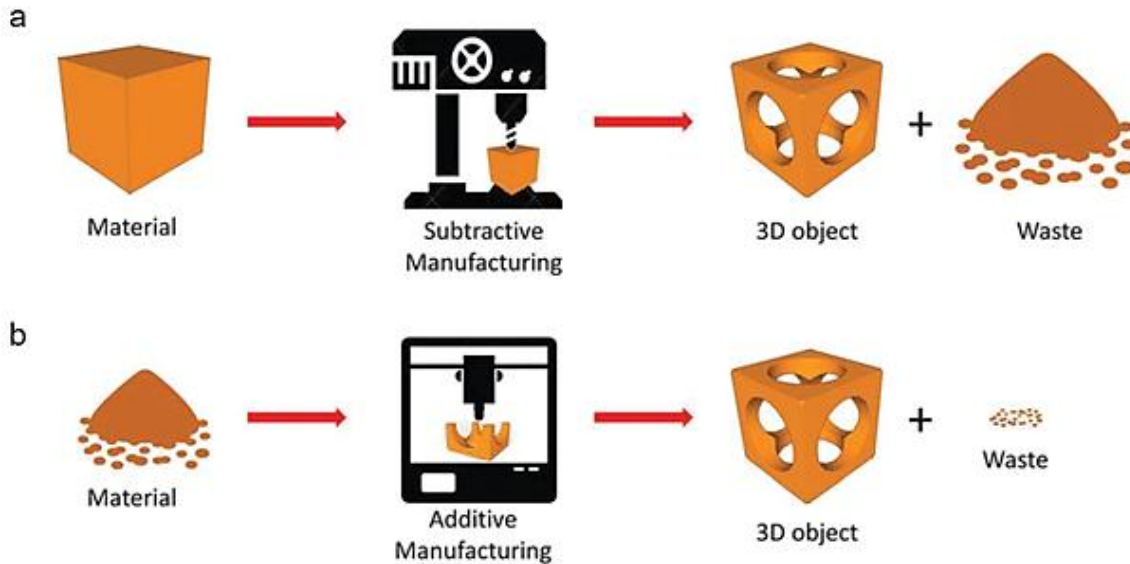
- Forms: Filament, Powder, Resin
- Processes & Materials:
 - FDM: ABS, PLA, PETG, Nylon, PEEK – widely used for ease of printing and affordability.
 - SLA/DLP: Acrylic and Epoxy based Photopolymer resins – suitable for high-resolution models with smooth surface finish.

- SLS: Nylon (PA12), Thermoplastic elastomers, Polyurethane – preferred for strong, flexible functional parts.
 - Applications: Prototypes, consumer products, medical models, functional tools, and low-volume end-use parts.
- 2. Metals: Metals in AM are valued for high strength, durability, and ability to replace conventionally machined parts.
 - Forms: Powder, Wire
 - Processes & Materials:
 - DMLS, LENS, DMD: Stainless Steel, Aluminium, Titanium, Inconel, Co-Cr alloys.
 - WAAM: Titanium, Aluminium, Steel – cost-effective for large-scale components.
 - Applications: Aerospace (lightweight structural parts), automotive (engine components), medical implants (titanium alloys), tooling, and defence equipment.
- 3. Ceramics: Ceramics offer biocompatibility and excellent thermal resistance, making them essential in medical and high-heat environments.
 - Forms: Powder
 - Processes & Materials:
 - Binder Jetting: Alumina, Zirconia, Silicon Carbide, Hydroxyapatite.
 - Applications: Dental crowns and bridges, orthopaedic implants, high-temperature components, electronics, and thermal barrier coatings.
- 4. Composites: Composites combine strength and lightweight properties, enabling high-performance applications.
 - Forms: Filament, Sheet
 - Processes & Materials:
 - FDM: Polymer filaments reinforced with Carbon fiber, Glass fiber and Kevlar.
 - CBAM: Metal or ceramic matrices reinforced with fibers.
 - Applications: Lightweight structural parts in aerospace and automotive industries, sporting goods, robotics, and defence.
- 5. Biomaterials: Biomaterials are crucial for bioprinting and healthcare innovations, focusing on biocompatibility and tissue integration.
 - Forms: Hydrogels, Biopolymers (filament & powder), Bio-inks
 - Processes & Materials:
 - SLA/DLP, SLS, FDM: Collagen, Gelatine, Alginate, Chitosan, PEG, PLA blends.
 - Applications: Tissue engineering scaffolds, artificial organs, drug delivery systems, prosthetics, and regenerative medicine.

1.1.7 ADDITIVE V/S CONVENTIONAL MANUFACTURING PROCESSES

Additive Manufacturing and Traditional (Subtractive) Manufacturing are two fundamentally different approaches to making parts. In AM, also known as 3D printing,

material is deposited layer by layer to build the product directly from a CAD model. This enables the creation of complex geometries, lightweight structures, and customized designs without the need for tooling. On the other hand, Subtractive Manufacturing is the traditional process where material is removed from a solid block through machining operations such as milling, drilling, or turning to achieve the desired shape. While AM offers design freedom and rapid prototyping, Subtractive Manufacturing provides high precision, smooth surface finish, and is well-suited for large-scale production. Together, these two approaches represent complementary methods in modern product development.



Differences between Additive and Subtractive Manufacturing

Aspect	Additive Manufacturing	Subtractive Manufacturing
Definition	Builds parts layer by layer by adding material.	Creates parts by removing material from a solid block (e.g., turning, milling, drilling).
Material Usage	Material-efficient (only deposits where needed, less waste).	High material wastage (chips, scrap from cutting).
Design Freedom	High – can produce complex geometries, lattices, internal channels.	Limited by tool access, cutting angles, and machine constraints.
Tooling Requirement	No special tools or moulds needed; directly from CAD.	Requires cutting tools, fixtures, jigs, and sometimes moulds.
Prototyping Speed	Rapid prototyping; easy design changes.	Slower prototyping; each change may need new tooling/setup.
Surface Finish	Generally rough, requires post-processing for smooth finish.	Very fine and smooth finish possible directly from machining.
Material Types	Polymers, metals, ceramics, composites	Almost all engineering materials (metals, plastics, wood, etc.).
Production Volume	Best for low-volume, customized, or complex parts.	Best for high-volume, standardized mass production.
Cost	Cost-effective for prototyping, but expensive for large batches.	Cost-effective for mass production.

1.1.8 APPLICATIONS OF ADDITIVE MANUFACTURING IN VARIOUS DOMAINS

1. Aerospace & Defence:

- Lightweight structural parts (brackets, housings).
- Complex fuel nozzles, turbine blades with internal cooling channels.
- Spare parts production on demand (reduces inventory).
e.g.: GE Aviation's 3D-printed fuel nozzle for LEAP engine.

2. Automotive:

- Rapid prototyping of car components.
- Custom tooling, jigs, and fixtures for assembly.
- Lightweight metal and plastic parts (interior panels, mounts, engine brackets).
e.g.: BMW, Ford, and Tesla use AM for prototyping and end-use parts.

3. Medical & Healthcare:

- Patient-specific implants (orthopaedic, dental, cranial).
- Customized prosthetics and orthotics.
- Surgical planning models (organ replicas, bone structures).
- Bioprinting of tissues and scaffolds (research stage).
e.g.: Titanium hip implants and dental crowns.

4. Consumer Products:

- Customized eyewear, footwear, and jewellery.
- Fashion and lifestyle products (bags, watches, apparel accessories).
- Toys, gadgets, and personalized household items.
e.g.: Adidas 4D shoes with 3D-printed midsoles.

5. Industrial & Manufacturing:

- Production of jigs, fixtures, and tooling inserts.
- Injection moulding inserts with conformal cooling channels.
- Spare parts supply chain optimization through on-demand printing.
e.g.: Siemens uses AM for turbine maintenance tools.

6. Architecture & Construction:

- 3D-printed houses, bridges, and structural components using concrete and composites.
- Scaled architectural models for design visualization.
- Sustainable construction using local raw materials.
e.g.: ICON's 3D-printed homes in the USA.

7. Education & Research:

- Hands-on prototyping for engineering and design students.
- Training for robotics, product design, and mechanical systems.
- Research into bioprinting, new materials, and hybrid processes.

8. Art, Jewellery & Fashion:

- Intricate jewellery designs with precious metals.
- Artistic sculptures and creative installations.
- Fashion accessories and custom designs.